

Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



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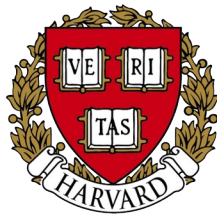
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Post-Doc
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*Statistics of the
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6 pub., 2 1st aut.

LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE



2018

2020

2025



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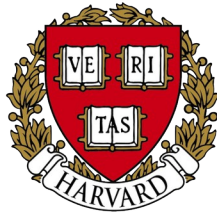
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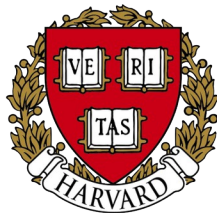
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Team Leader at CIML, Marseille
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Hybrid team, experimental and
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Ciml
IMMUNOLOGY

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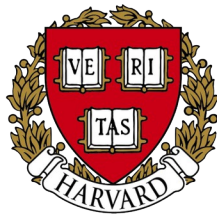


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CSS 2: Cancer &
Genetic diseases

Inserm

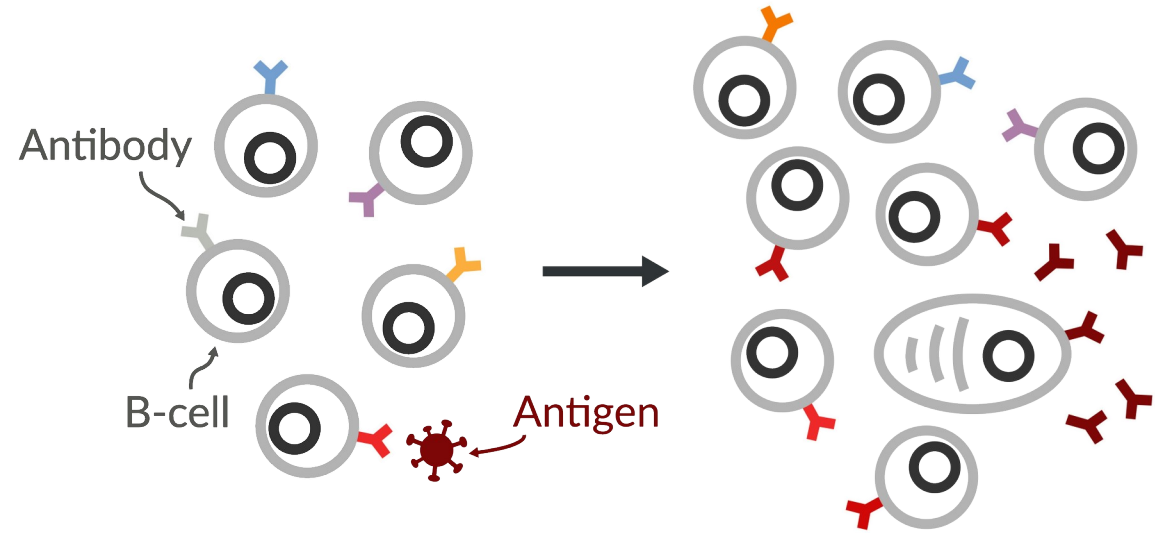


Immune repertoires

Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).

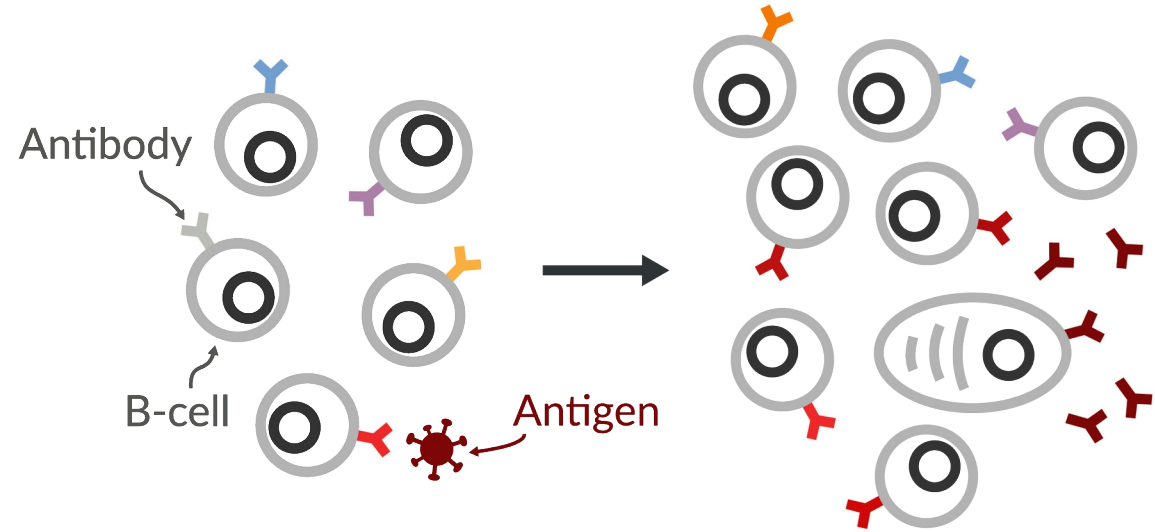
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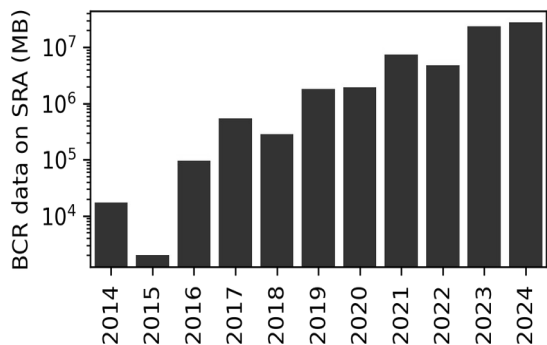
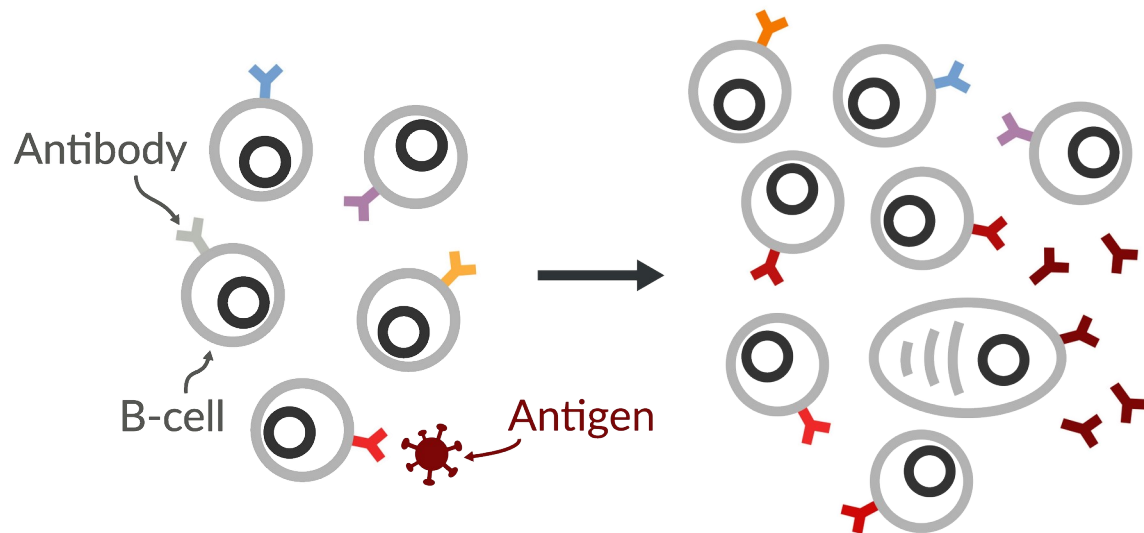
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The repertoire as a prognosis tool ?

Immune repertoires

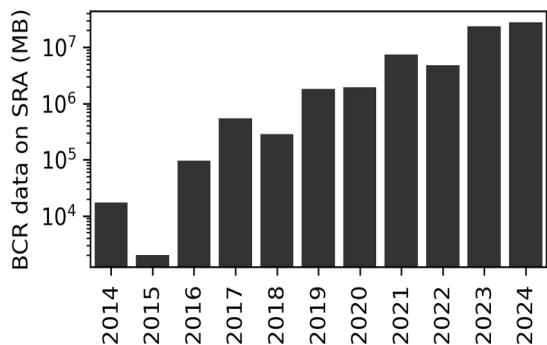
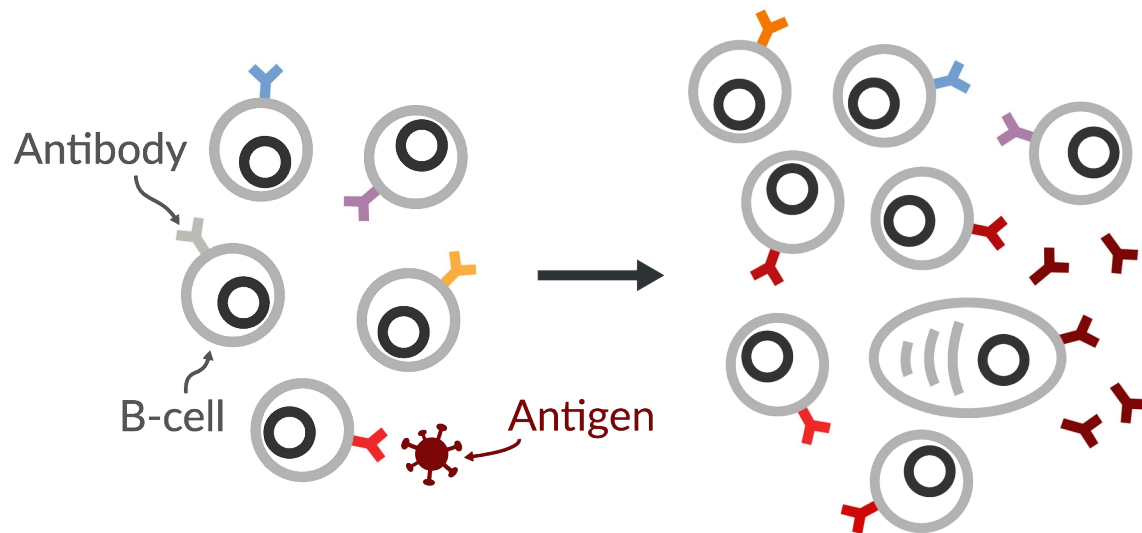
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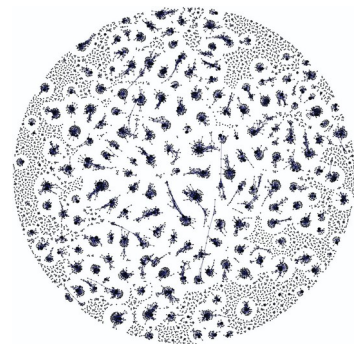
The repertoire as a prognosis tool ?
Easy to access...

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The repertoire as a prognosis tool ?
Easy to access... but hard to interpret.



Immune repertoires

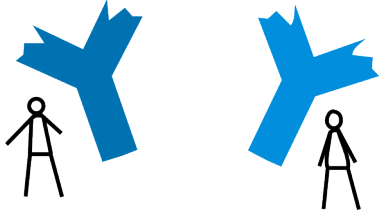
Statistical models, **generation**¹ & **selection**² of the receptors. **Dupic et al., PLOS Comp. Bio (2019)**

$$P(\text{Y}) = \dots$$

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Public repertoire: do shared receptors imply a shared response?

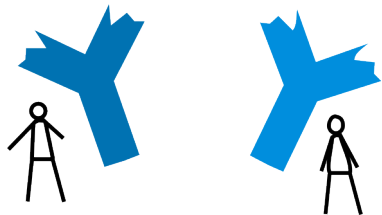
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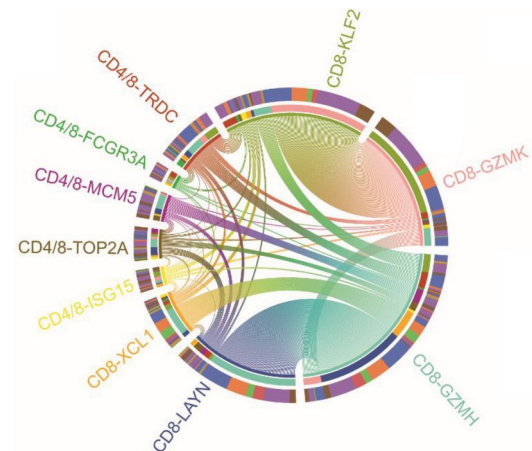
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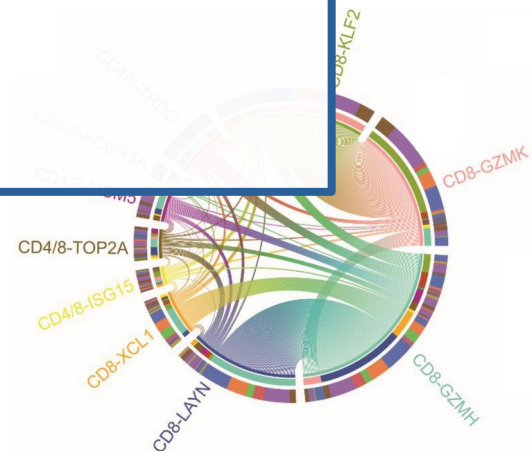
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shared Response?

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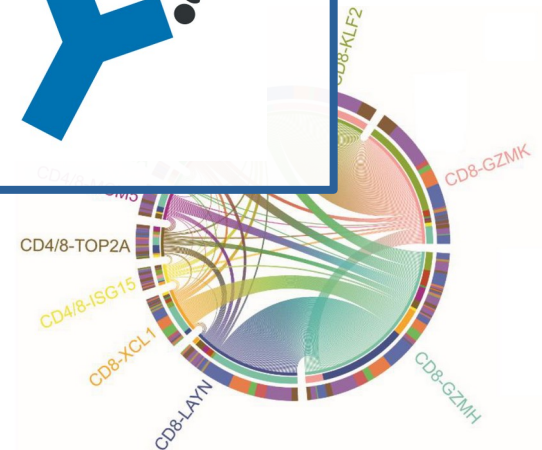
Current methods don't integrate the **functional properties** of the receptor.



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Abbate, Dupic

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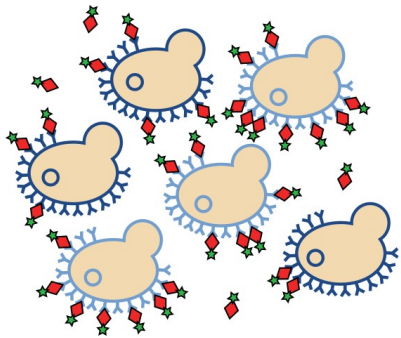
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



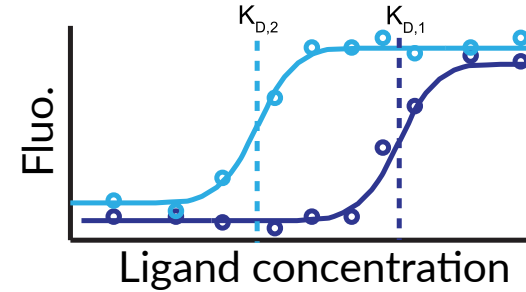
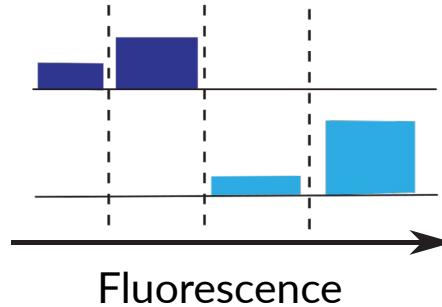
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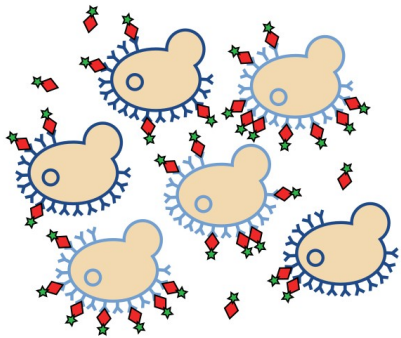
Expression on the yeast with fluorescent antigen

Cell sorting, sequencing and statistics



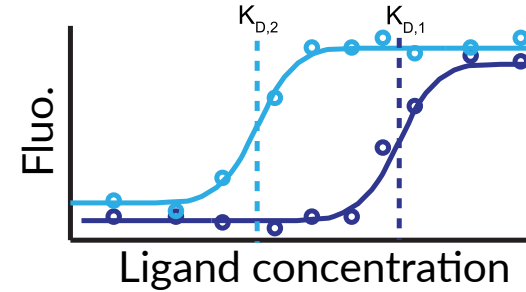
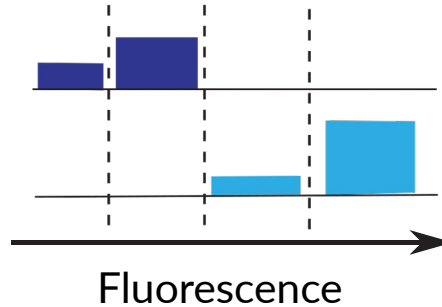
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Viral proteins

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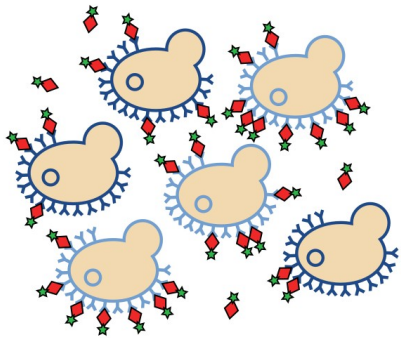
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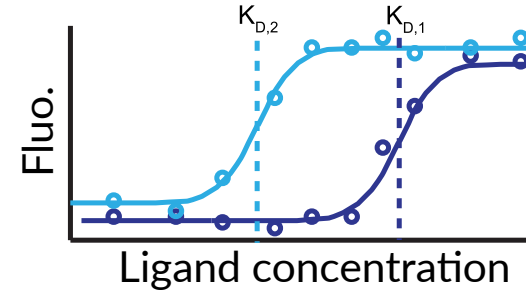
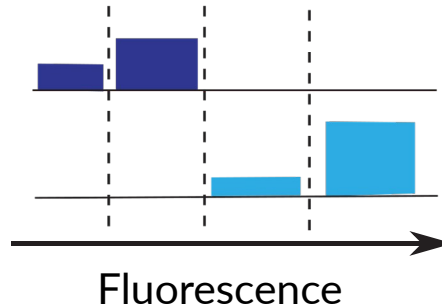
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Antibodies

Phillips, ..., Dupic et al., eLife (2021)

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Dupic, et al., Genetics (2026, in revision)



Measuring affinity at repertoire scale



Measuring affinity at repertoire scale

Long-term B-cell
response to SARS-
CoV-2 vaccination

Moulana, ..., Dupic*, Desai* (in preparation)



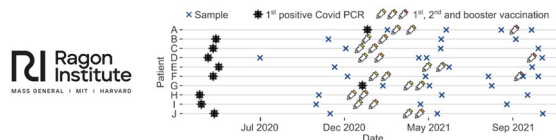
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Repertoire Sequencing →



Common BCRs →

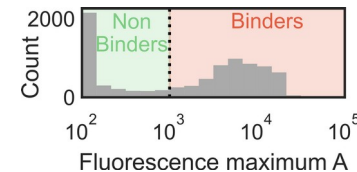


Affinity

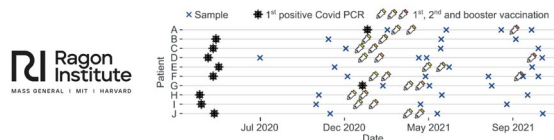
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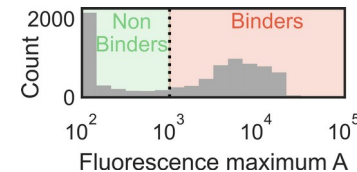
Affinity

1M BCRs tested $\Rightarrow$ 5000 binders

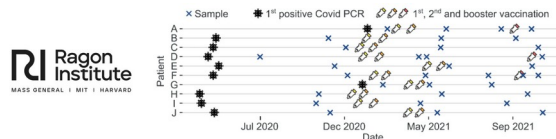
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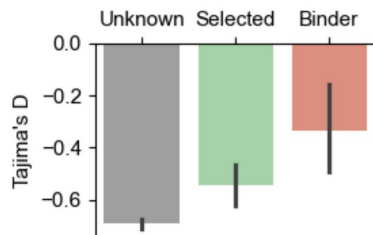
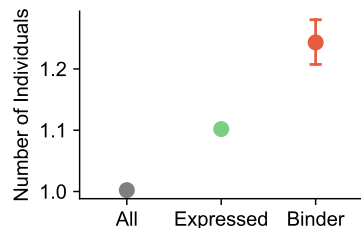
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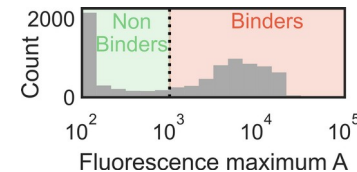


The repertoire statistics of **binders**
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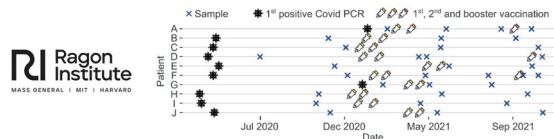
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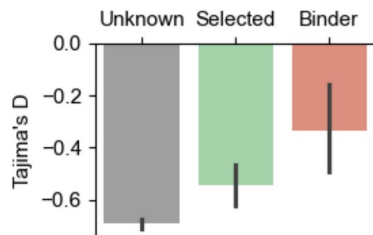
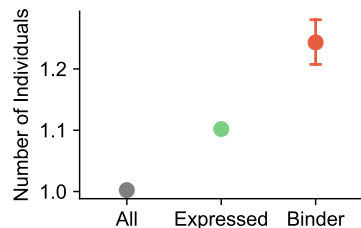
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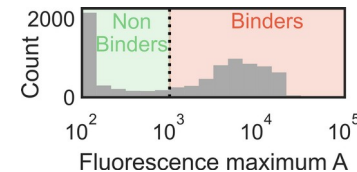
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Improving binding models with transfer learning

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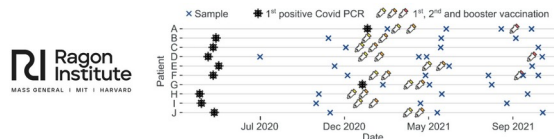


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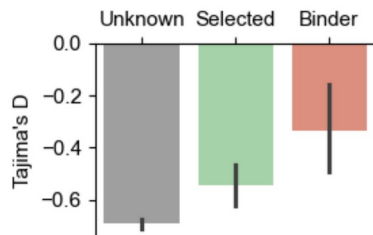
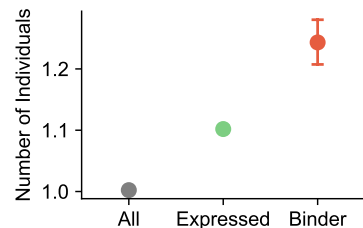
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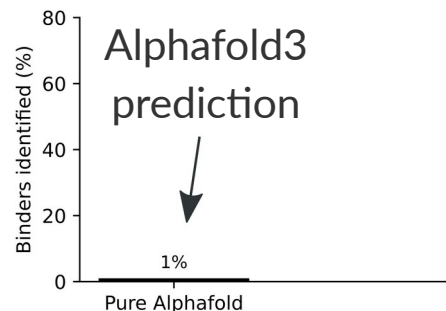
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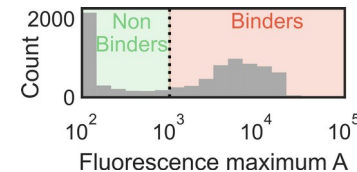
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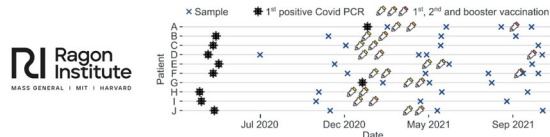


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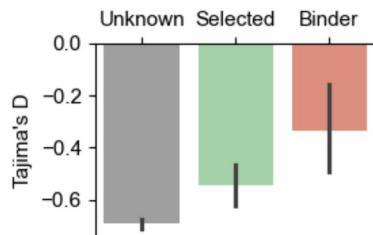
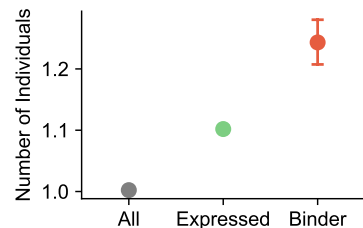
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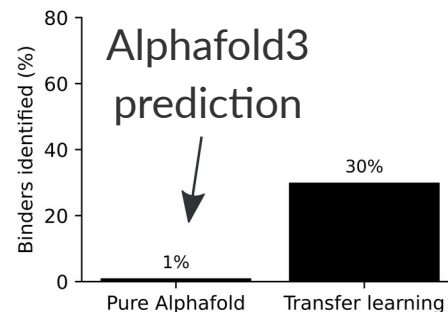
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AlphaFold3 embedding
 $\times$
Our measurements

Measuring affinity at repertoire scale



Select
antibodies of
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Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

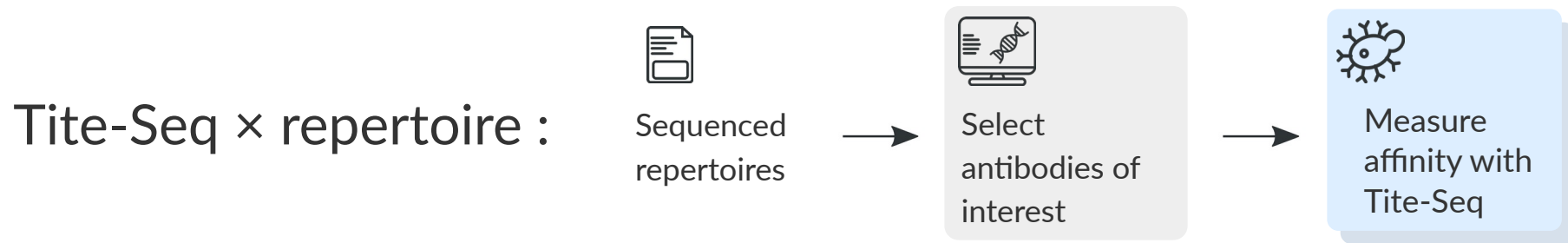


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Measure
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Measuring affinity at repertoire scale



Tite-Seq is precise, mature and cheap

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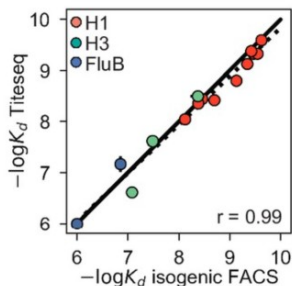


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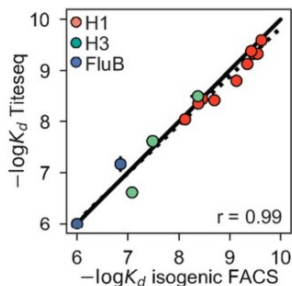


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Starr et al., Nature, 2021
Greaney et al., Cell Host & Microbes, 2023
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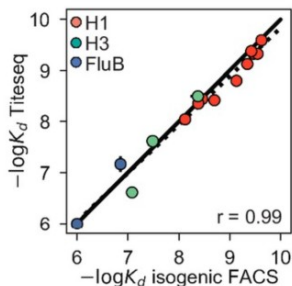


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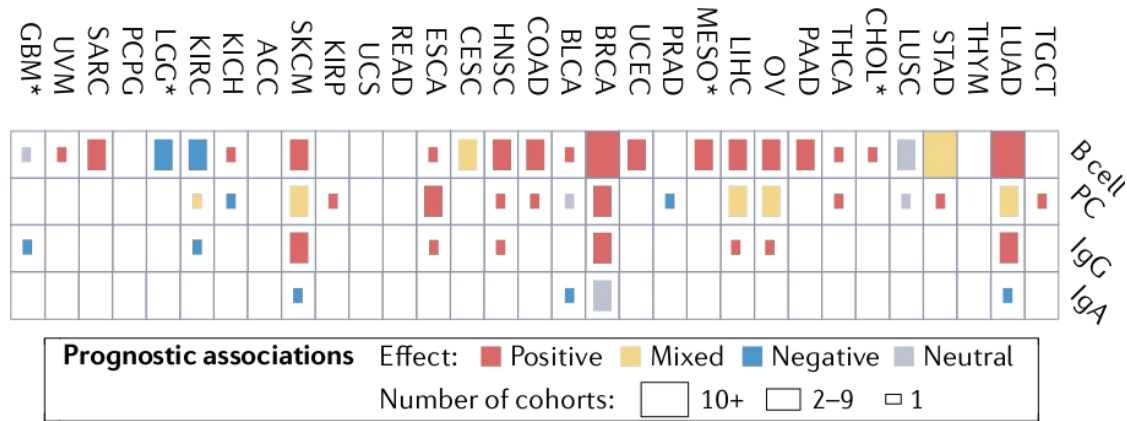
Tumor Infiltrated B-Lymphocytes (TIL-B)

Immunotherapy is a revolution, but is still ineffective for 80% of patients.

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TIL-Bs improve prognosis and response to immunotherapy

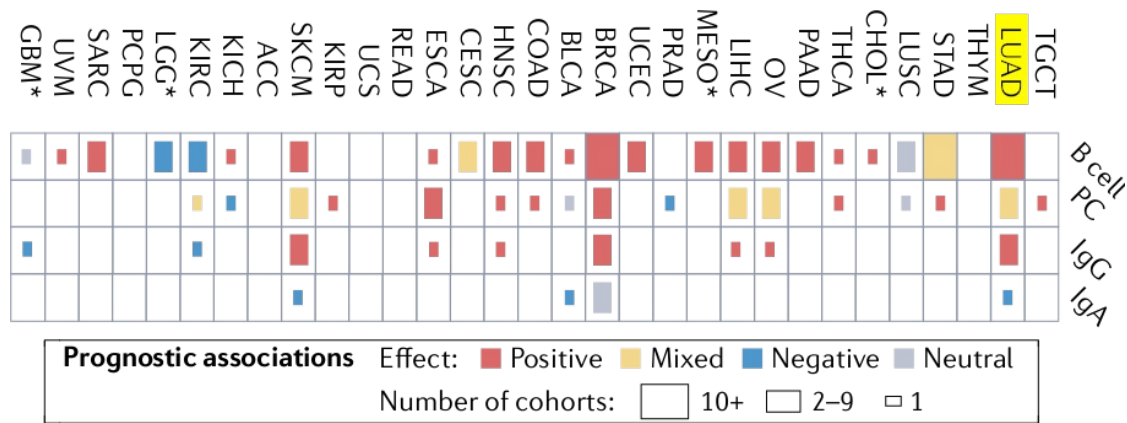


Adapted from *Laumont et al., Nat. Rev. Cancer (2022)*

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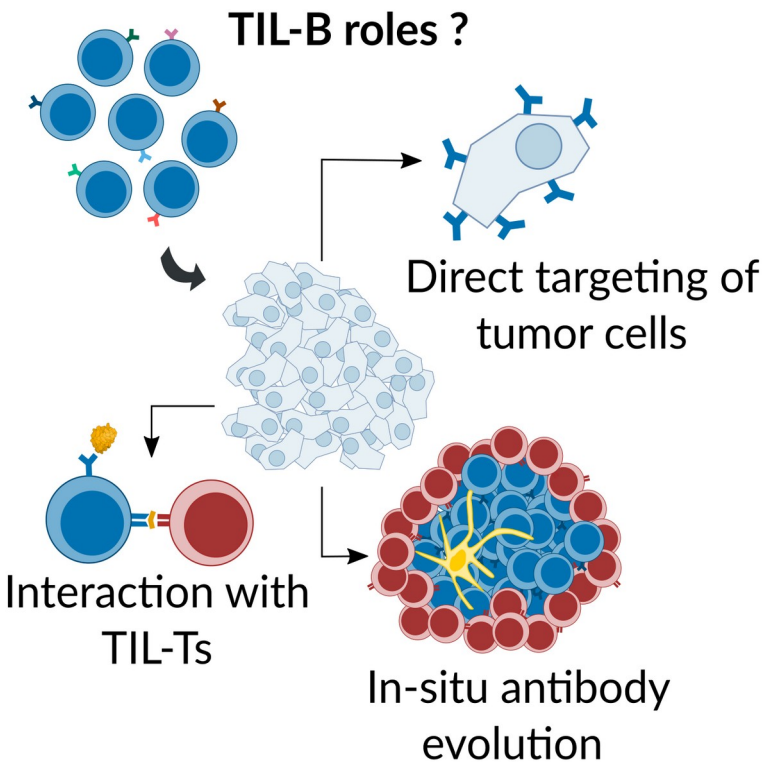
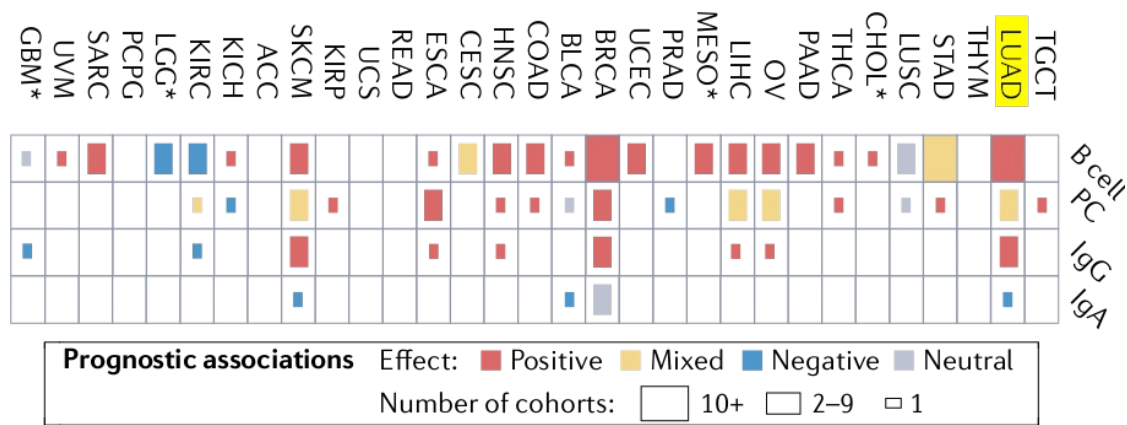


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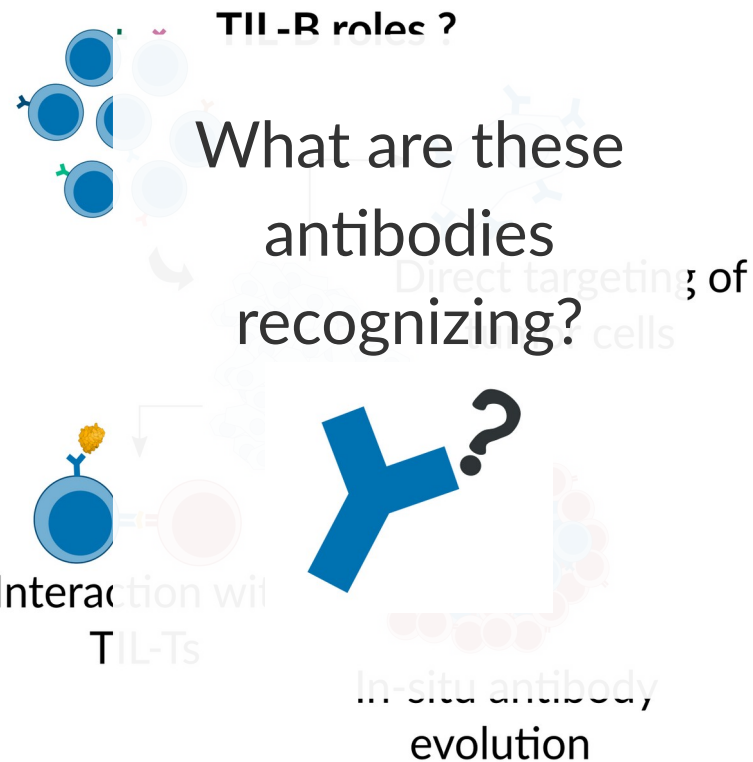
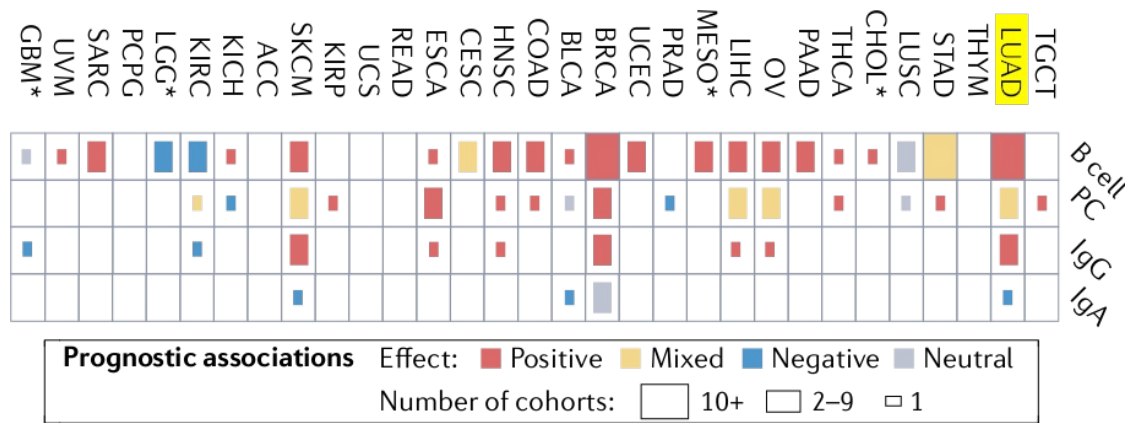


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The public TIL-B repertoire

Only **70 pairs** antigen /
TIL-antibody identified.

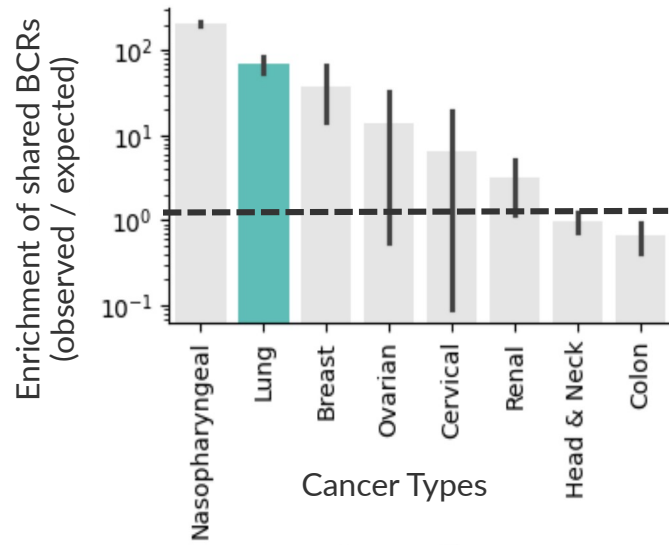
Laumont et al., Nat. Rev. Cancer (2022)

The public TIL-B repertoire

Only 70 pairs antigen /
TIL-antibody identified.

Laumont et al., Nat. Rev. Cancer (2022)

The intra-tumoral repertoire is
public $\Rightarrow$ directed response !



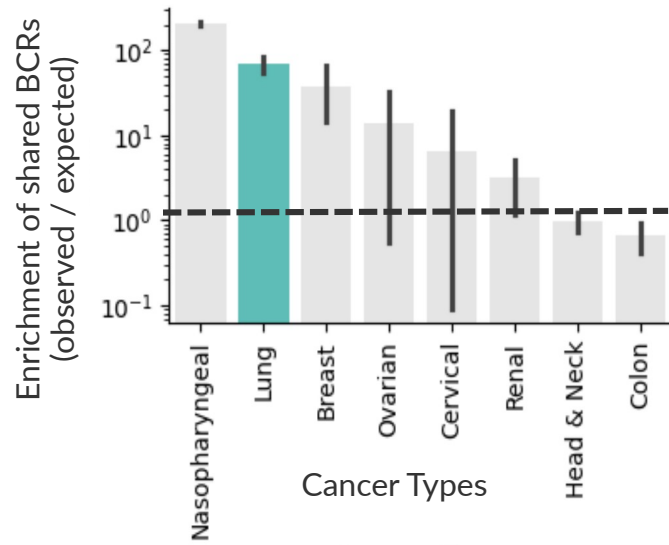
Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

The public TIL-B repertoire

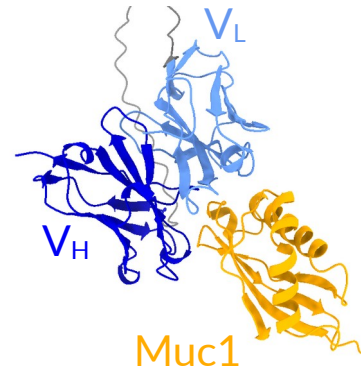
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Preliminary Data

Lung cancer:

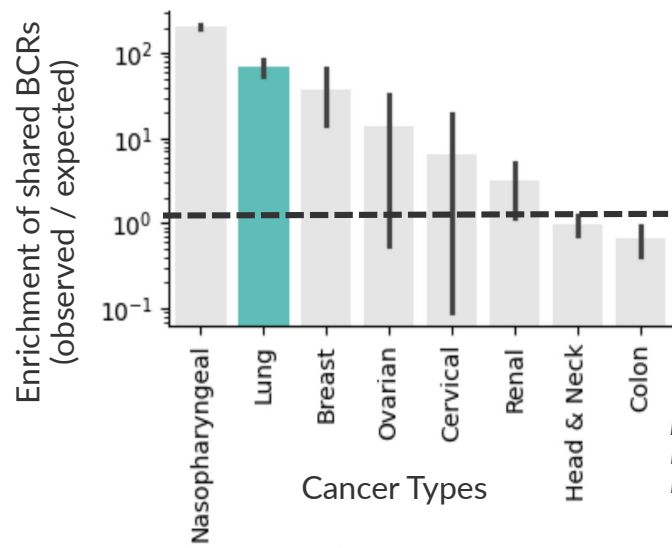
500 BCRs tested
in-silico against 5
antigens

The public TIL-B repertoire

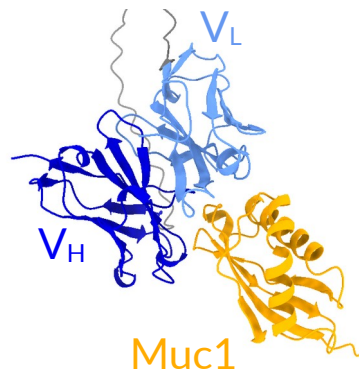
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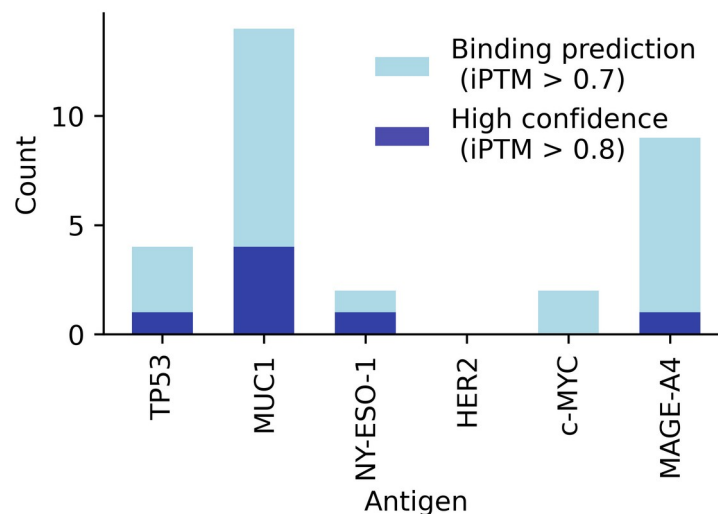
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Preliminary Data

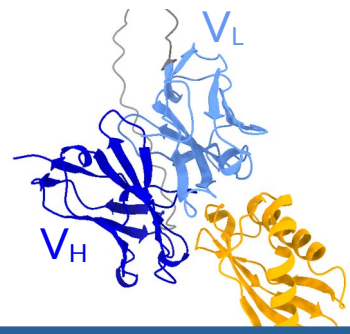
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The public TIL-B repertoire

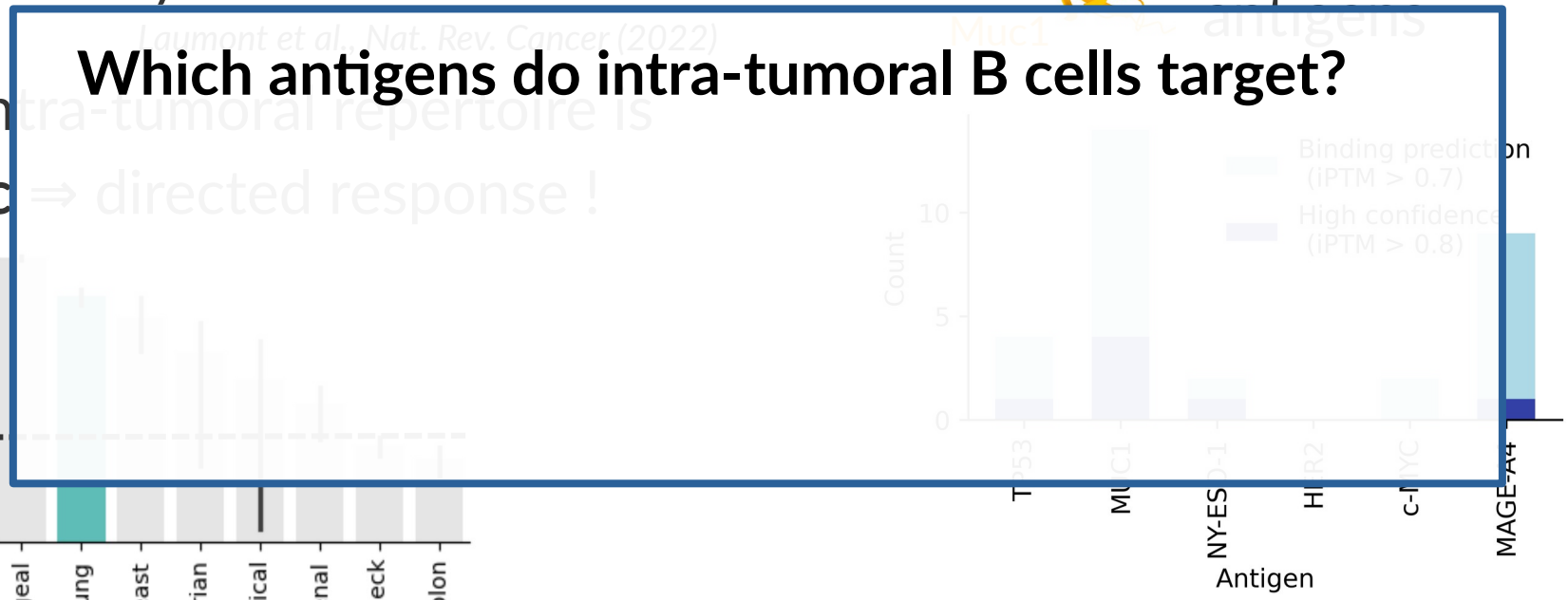
Only 70 pairs antigen /
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Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

The intra-tumoral repertoire is
public



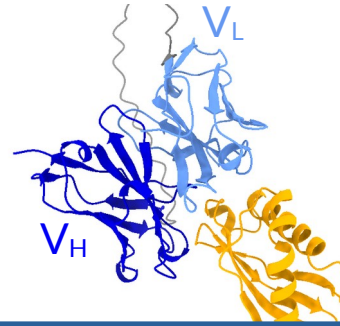
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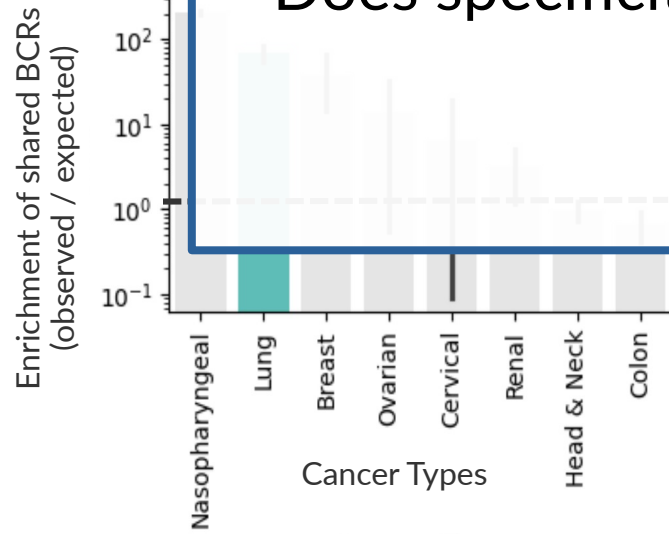
Preliminary Data

Lung cancer:
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Which antigens do intra-tumoral B cells target?

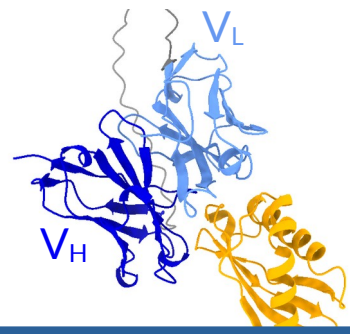
- Do plasma cells have different specificity?
- Does specificity change during affinity maturation?



Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

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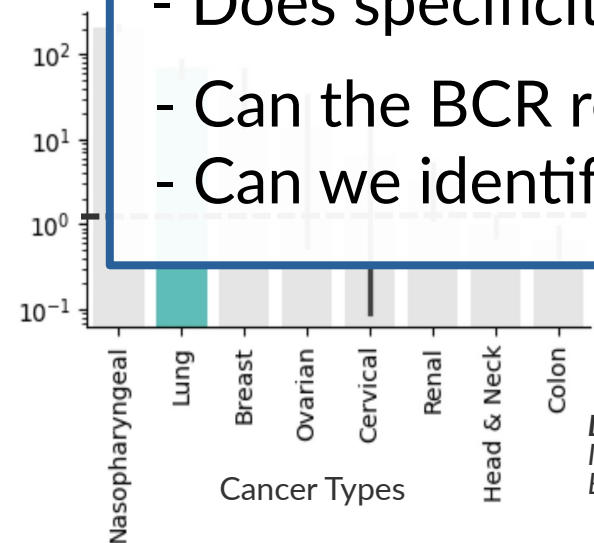


Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

- Do plasma cells have different specificity?
- Does specificity change during affinity maturation?
- Can the BCR repertoire serve as a **biomarker**?
- Can we identify new **targetable antigens**?

Enrichment of shared BCRs
(observed / expected)



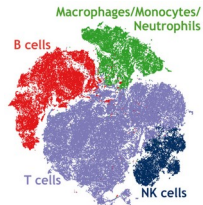
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Which antigens are targeted by TIL-Bs



Non-small cell lung
cancer dataset(s)



Public data

Hao (2022)

Prazanowska (2023)

216 patients

(PBMC samples)

LUCA-PI

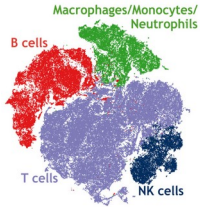
900 patients

(tumor & PBMC samples)

Which antigens are targeted by TIL-Bs



Non-small cell lung cancer dataset(s)



Public data

Hao (2022)

Prazanowska (2023)



216 patients

(PBMC samples)

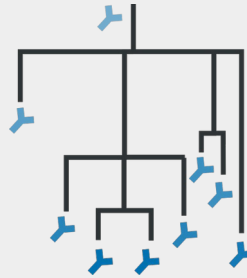
LUCA-PI

900 patients

(tumor & PBMC samples)



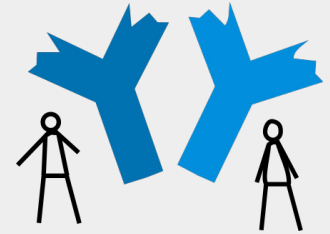
Identify TIL-B of interest *in-silico* (WP2)



Clonality & transcriptomics

Plasma / Memory B-cells.

Clonal lineages under selection



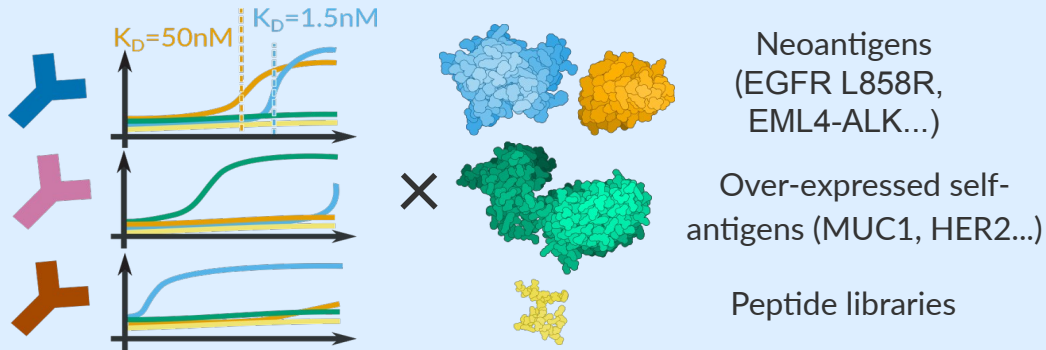
Comparisons between patients

Identify similar receptors between repertoires

Which antigens are targeted by TIL-Bs



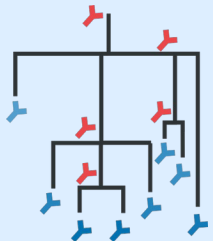
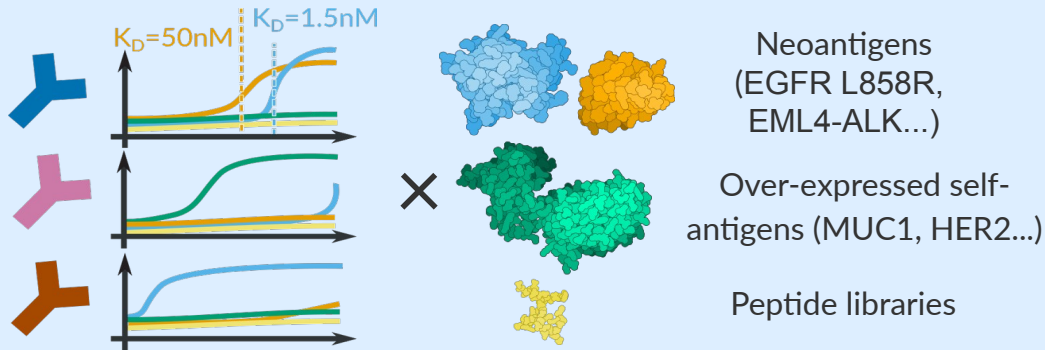
Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)

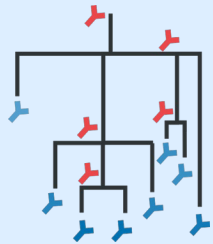
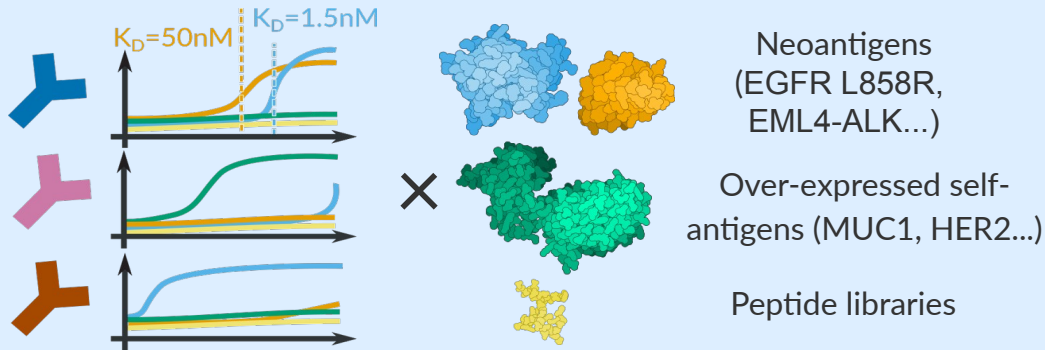


Ancestral
reconstruction
Did the antibody evolve
against this antigen ?

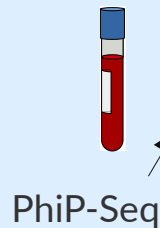
Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Ancestral reconstruction
Did the antibody evolve against this antigen ?

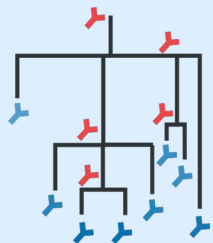
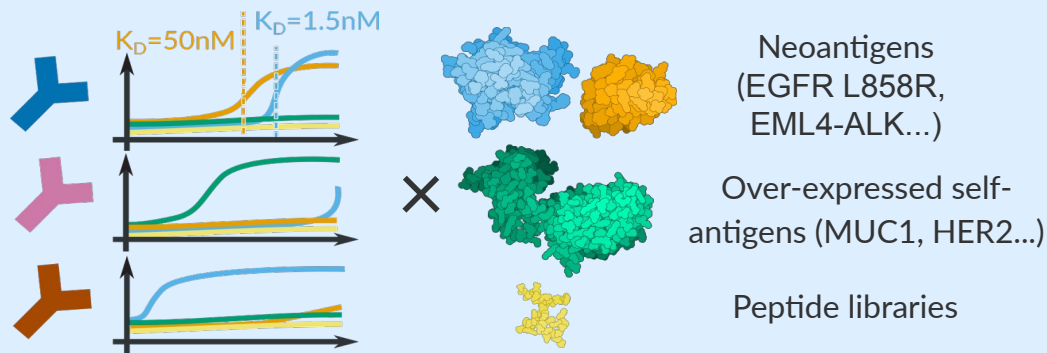


Validation,
sera / PBMC
Pioneer / LUCA-
PI Cohort

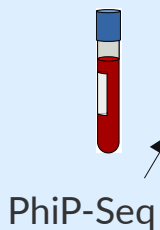
Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Ancestral reconstruction
Did the antibody evolve against this antigen?

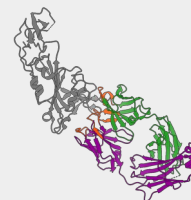


Validation,
sera / PBMC
Pioneer / LUCA-PI Cohort



Models & learning (WP2)

Generalize Ab-Ag predictions



Integrate specificity to prognosis



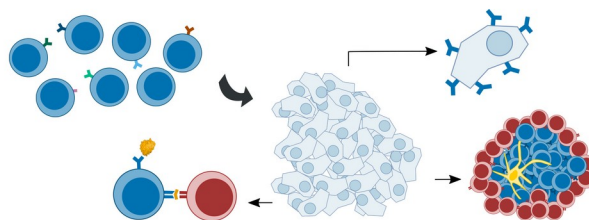
?

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

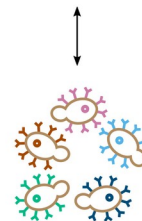


Project summary

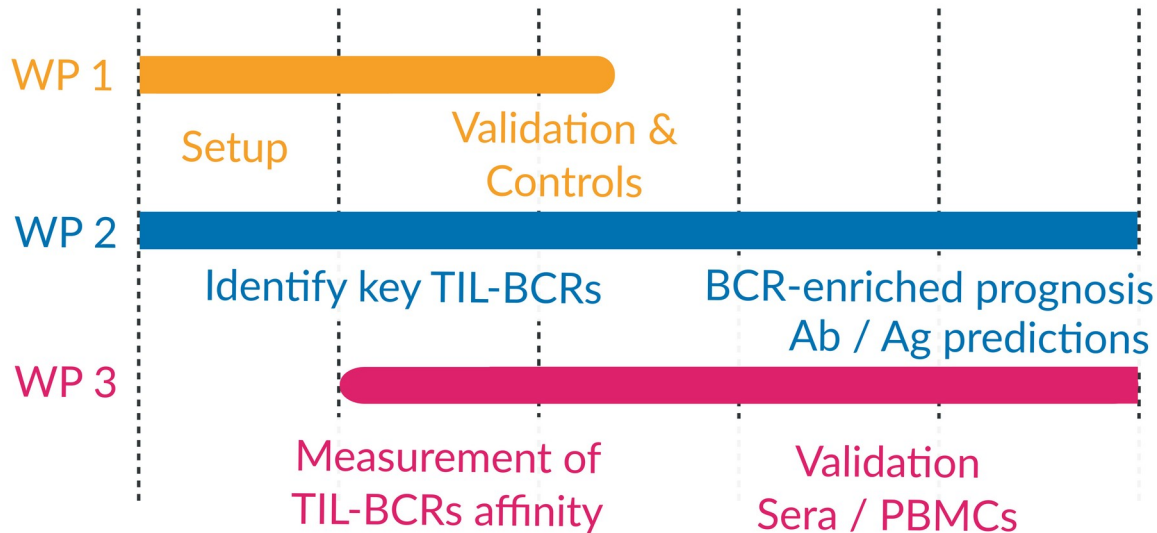


Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity



Timeline



Post-doctoral researcher

Partner institutions

Inserm

amU
Aix Marseille Université

Co-financing
Idex $\Rightarrow$ 1 PhD

ap. hm

Cimi
IMMUNOLOGY

Team leader
Equipment,
lab, technician

Collaborations

CRCM
Centre de Recherche
en Cancérologie de Marseille

UCSF
 LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE

P. Milpied
Q. Marcou
A. Phillips
T. Mora, A. Walczak

LUCA-PI

THE PIONEER PROJECT